

Introduction to Nonparametric Statistics
Tutorial 2 : Estimation of the Distribution Function

Exercise 1 :

Let $X_1, \dots, X_n \stackrel{iid}{\sim} Be(p)$. We want to test, using a non-asymptotic test,

$$H_0 : p = 1/2 \quad \text{against} \quad p \neq 1/2.$$

1. Recall the testing procedure seen in Exercise 5 of Tutorial 1.
2. Give the expression of the p -value.

Exercise 2 : Consider an i.i.d. sample $X_1^n = (X_1, \dots, X_n)$. Let F denote its cumulative distribution function and $\hat{F}_n$ the empirical distribution function associated with this sample. Let F_0 be a given cumulative distribution function.

1. Show that if F_0 is continuous, the distribution under H_0 of the statistic

$$h_n^+(X, F_0) = \sup_{t \in \mathbb{R}} \left\{ \hat{F}_n(t) - F_0(t) \right\}_+$$

does not depend on F_0 .

2. Propose a testing procedure for

$$H_0 : F = F_0 \quad \text{against} \quad H_1 : \exists t \in \mathbb{R} \text{ such that } F(t) > F_0(t).$$

Exercise 3 : In this exercise we study the power of the Kolmogorov–Smirnov test. We consider an i.i.d. sample $(X_1, \dots, X_n)$ with cdf F and empirical cdf $\hat{F}_n$. We want to test

$$H_0 : F = F_0 \quad \text{against} \quad H_1 : F \neq F_0$$

where F_0 is a given distribution. We want to know whether the test is able, with high probability, to detect that the sample does not follow the distribution F_0 when this is indeed the case, provided that the sample size is large enough. In other words, we want to know whether the test is powerful.

We assume that F_0 is continuous.

1. Using the DKW inequality seen in class, show that the $(1 - \alpha)$ -quantile $\xi_{n,1-\alpha}$ of the Kolmogorov–Smirnov statistic satisfies $\xi_{n,1-\alpha} = O(\frac{1}{\sqrt{n}})$ as $n \rightarrow \infty$.
2. Assume that $F \neq F_0$, i.e., the sample does not follow the distribution F_0 . Show that if we set

$$\underline{\beta}(F) = \mathbb{P}_{X_1, \dots, X_n \stackrel{iid}{\sim} F} (\|\hat{F}_n - F_0\|_\infty \geq \xi_{n,1-\alpha}),$$

then

$$\underline{\beta}(F) \xrightarrow{n \rightarrow \infty} 1.$$

Exercise 4 : Consider an i.i.d. sample $X_1^n = (X_1, \dots, X_n)$ of random variables. Let F denote the cdf and $\hat{F}_n$ the empirical cdf associated with this sample. If the variables X_i are normally distributed with parameters μ and σ^2 , we also denote by N_{μ, σ^2} their common cdf.

1. Assume that $F = N_{\mu, \sigma^2}$. Determine the maximum likelihood estimator $(\hat{\mu}, \hat{\sigma}^2)$ of (μ, σ^2) .
2. Define

$$\Delta_n = \sup_{t \in \mathbb{R}} |\hat{F}_n(t) - N_{\hat{\mu}, \hat{\sigma}^2}(t)|.$$

Show that if $F = N_{\mu, \sigma^2}$, then the distribution of Δ_n does not depend on μ and σ^2 .

3. Deduce a test for membership in the family of normal distributions, i.e., a test of

$$H_0 : F \in \mathcal{FN} \quad \text{against} \quad H_1 : F \notin \mathcal{FN},$$

where

$$\mathcal{FN} = \{G : \exists (\mu, \sigma^2) \in (\mathbb{R} \times \mathbb{R}_+^*) \text{ tel que } G = N_{\mu, \sigma^2}\}.$$

4. *Application (largely independent of the rest of the exercise) :* The distribution of the test statistic in Question 3 has been tabulated. We also consider the test, seen in class (Section 2.3.2 of the course notes), for membership in the exponential family. We provide below some useful quantiles for $n = 4$:

	$q_{5\%}$	$q_{10\%}$	$q_{90\%}$	$q_{95\%}$
Normal-family membership test statistic	0.18	0.20	0.36	0.39
Exponential-family membership test statistic	0.21	0.23	0.44	0.48

Consider the observed sample of size $n = 4$:

$$0.66 \quad 3.51 \quad 1.92 \quad 1.05$$

We want to test whether this sample is distributed according to a normal distribution and whether it is distributed according to an exponential distribution. To do so, we propose to apply the test constructed above and the test from the course notes. In Figure 1 (left and right) we have plotted the empirical cdf corresponding to the given sample. Moreover, on the left we have plotted the function $N_{\hat{\mu}, \hat{\sigma}^2}$, where $\hat{\mu}$ and $\hat{\sigma}^2$ are the maximum likelihood estimators ($\hat{\mu} = 1.78$ $\hat{\sigma}^2 = 1.20$). On the right we have plotted the cdf of the exponential distribution with parameter $\hat{\lambda}$ ($\hat{\lambda} = 0.56$).

- (a) By reading Figure 1, give the value of the test statistics for the two tests.
- (b) Using the quantiles given above, perform the two tests at level 5%.
- (c) Do the two conclusions seem consistent to you ?

Exercise 5 : Let $X_1, \dots, X_n$ be an i.i.d. sample with continuous cdf F . Let $p \in]0, 1[$. We denote by x_p the quantile of order p of F .

In this exercise we show how to obtain an asymptotic confidence interval for x_p without knowing F (assuming only that F is continuous).

Let $\alpha \in]0, 1[$ be a confidence level. Let $\hat{F}_n$ be the empirical cdf and let $\hat{F}_n^{(-1)}$ be the quantile function (generalized inverse) associated with $\hat{F}_n$.

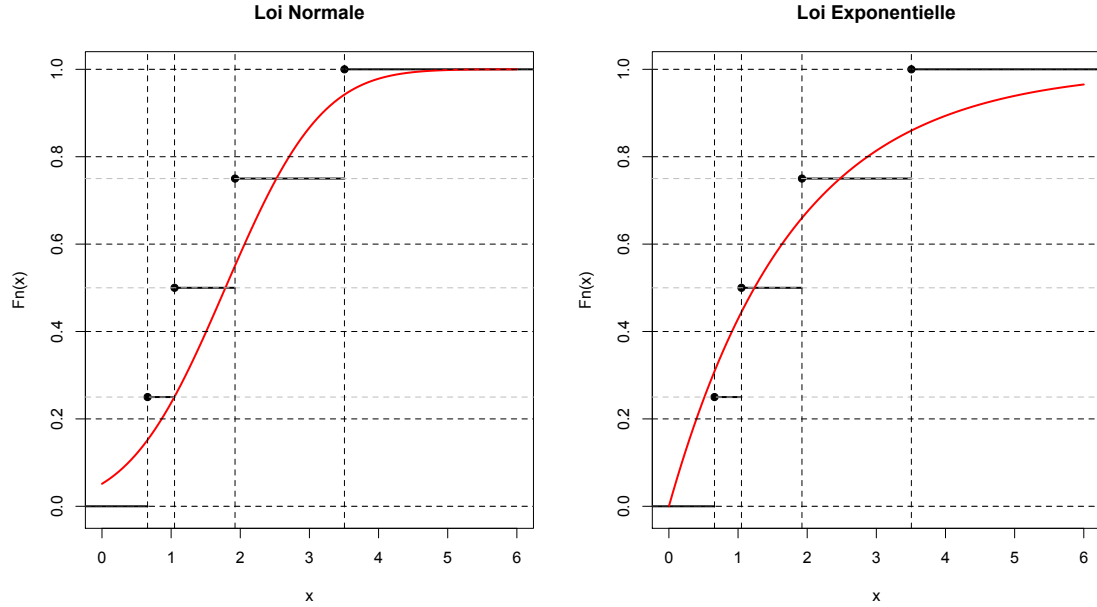


FIGURE 1 – Exercise 4 : empirical cdf of the sample (step function) and the cdf being tested.

1. Give the limit in distribution of $\sqrt{n}(\hat{F}_n(x) - F(x))$ as $n \rightarrow \infty$.
2. Give the limit in distribution of $\sqrt{n}(\hat{F}_n(x_p) - p)$ as $n \rightarrow \infty$.
3. Deduce that

$$P\left(p - \Phi^{-1}(1 - \alpha/2)\sqrt{\frac{p(1-p)}{n}} \leq \hat{F}_n(x_p) < p + \Phi^{-1}(1 - \alpha/2)\sqrt{\frac{p(1-p)}{n}}\right) \xrightarrow{n \rightarrow +\infty} 1 - \alpha$$

where Φ is the cdf of the standard normal distribution.

4. Using the generalized inverse function $\hat{F}_n^{(-1)}$, deduce an asymptotic confidence interval of level $1 - \alpha$ for x_p .

Exercise 6 : The goal of this exercise is to propose a multiple-testing procedure when the number of hypotheses to test is large. Throughout the exercise we consider a statistical model $(\Omega, \mathcal{A}, \mathbb{P}_\theta, \theta \in \Theta)$.

Part A. We first consider the simple setting where we want to test

$$H_0 : \theta = \theta_0 \quad \text{against} \quad H_1 : \theta \in \Theta_1$$

where $\theta_0 \notin \Theta_1$. We observe a real-valued random variable X with distribution $\mathbb{P}_\theta$. For a given $\alpha \in]0, 1[$, we consider a test of the form $\varphi_\alpha(X) = 1_{X \geq k_\alpha}$ where $k_\alpha \in \mathbb{R}$. Let F_θ be the cdf of X under $\mathbb{P}_\theta$. We assume that F_θ is continuous.

1. Give the expression of the observed p -value in terms of F_{θ_0} .

2. What is $P_{\theta_0}(p(X) \leq u)$ for $u \in]0, 1[$?

Remark : to understand the more general case where the null hypothesis is composite, see the 2014 exam.

Part B. In this part (independent of Part A), for $m \in \mathbb{N}^*$ we consider $2m$ subsets of Θ denoted $\Theta_{01}, \Theta_{11}, \Theta_{02}, \Theta_{12}, \dots, \Theta_{0m}, \Theta_{1m}$ such that for every $i \in \{1, \dots, m\}$,

$$\Theta_{0i} \cap \Theta_{1i} = \emptyset,$$

and we want to perform **simultaneously** m tests

$$H_{0i} : \theta \in \Theta_{0i} \quad \text{against} \quad H_{1i} : \theta \in \Theta_{1i}, \quad i = 1, \dots, m.$$

For simplicity, we assume that the null hypotheses are singletons $\Theta_{0i} = \{\theta_{0i}\}$ and that the individual tests are of size α . Let I_0 be the set of indices i for which H_{0i} is true :

$$I_0 = \{i \in \{1, \dots, m\} : H_{0i} \text{ is true}\}.$$

We want to construct a multiple-testing procedure that returns a set $\hat{R} \subset \{1, \dots, m\}$ corresponding to the indices i for which H_{0i} is rejected.

Remark : failing to account for the fact that one is testing many hypotheses simultaneously is part of what is known as “p-hacking”; see for instance <https://statisticsbyjim.com/hypothesis-testing/p-hacking/>, Section “Excessive Hypothesis Testing and Multiple Comparisons”.

We denote by FP the number of indices corresponding to null hypotheses incorrectly rejected, and by TP the number of indices corresponding to null hypotheses correctly rejected :

$$\text{FP} = \text{card}(\hat{R} \cap I_0), \quad \text{TP} = \text{card}(\hat{R} \setminus I_0).$$

FP is the number of *false positives* and TP the number of *true positives*. Ideally, we want a procedure such that FP is small and TP is large. Let $\hat{p}_i$ be the p -value of the test of H_{0i} against H_{1i} . Thus, $\hat{p}_i$ is a statistic satisfying, for all $u \in]0, 1[$,

$$\mathbb{P}_{\theta_{0i}}(\hat{p}_i \leq u) = u. \tag{1}$$

1. We first propose the Bonferroni procedure, which controls FP by setting, for $\alpha \in]0, 1[$:

$$\hat{R} = \left\{ i \in \{1, \dots, m\} : \hat{p}_i \leq \frac{\alpha}{m} \right\}.$$

(a) Show that

$$\mathbb{P}(\text{FP} > 0) \leq \sum_{i \in I_0} \mathbb{P}_{\theta_{0i}}(\hat{p}_i \leq \alpha/m).$$

(b) Using (1), deduce that

$$\mathbb{P}(\text{FP} > 0) \leq \alpha.$$

2. The Bonferroni procedure controls the number of false positives but may produce too few true positives. In other words, it is overly conservative. We therefore consider the following alternative. Let $f : \{0, 1, \dots, m\} \rightarrow [0, m]$ be a nondecreasing function, and order the statistics $\hat{p}_i$ in increasing order :

$$\hat{p}_{(1)} \leq \hat{p}_{(2)} \leq \dots \leq \hat{p}_{(m)}.$$

We seek to control the ratio FDR defined by

$$\text{FDR} = \mathbb{E} \left[\frac{\text{FP}}{\text{FP} + \text{TP}} 1_{\{\text{FP} + \text{TP} \geq 1\}} \right]$$

with the convention $0/0 = 0$.

We define, for $\alpha \in]0, 1[$,

$$\hat{R} = \left\{ i \in \{1, \dots, m\} : \hat{p}_i \leq \frac{\alpha f(\hat{k})}{m} \right\}$$

with

$$\hat{k} = \max \left\{ k \in \{1, \dots, m\} : \hat{p}_{(k)} \leq \frac{\alpha f(k)}{m} \right\}.$$

In particular, we set $\hat{k} = 0$ and $\hat{R} = \emptyset$ if for every integer k , $\hat{p}_{(k)} > \frac{\alpha f(k)}{m}$.

- (a) Show that

$$\hat{k} = \text{card}(\hat{R})$$

and that for $j \geq \hat{k}$,

$$f(\hat{k}) \leq f(\min(j, m)).$$

- (b) Establish that

$$\text{FDR} = \sum_{i \in I_0} \mathbb{E} \left[1_{\{\hat{p}_i \leq \alpha f(\hat{k})/m\}} \times \frac{1_{\{\hat{k} \geq 1\}}}{\hat{k}} \right].$$

- (c) Show that if $\hat{k} \geq 1$,

$$\frac{1}{\hat{k}} = \sum_{j=1}^{+\infty} \frac{1_{\{j \geq \hat{k}\}}}{j(j+1)}.$$

- (d) Deduce that

$$\text{FDR} \leq \frac{\alpha \text{card}(I_0)}{m} \sum_{j=1}^{+\infty} \frac{f(\min(j, m))}{j(j+1)}.$$

- (e) Conclude that if f satisfies

$$\sum_{j=1}^{+\infty} \frac{f(\min(j, m))}{j(j+1)} \leq 1 \tag{2}$$

then

$$\text{FDR} \leq \alpha.$$

(f) Give an example of a function f satisfying (2).

Remark : these results can also be extended to the case of composite null hypotheses (see the 2014 exam).

Remark : if the m tests are independent, one can in fact take f equal to the identity ; then one obtains more true positives than in the previous case while still having an FDR bounded by α . The resulting procedure is :

$$\hat{R} = \left\{ i \in \{1, \dots, m\} : \hat{p}_i \leq \frac{\alpha \hat{k}}{m} \right\}$$

with

$$\hat{k} = \max \left\{ k \in \{1, \dots, m\} : \hat{p}_{(k)} \leq \frac{\alpha k}{m} \right\}.$$

This is the Benjamini–Hochberg procedure (see the 2018 midterm).

R code : if the m independent test p -values are stored in the vector $\mathbf{p}$, then the following code computes $\hat{R}$, the set of indices rejected by the Benjamini–Hochberg procedure when we want an FDR smaller than 5% :

```
k<-sum(sort(p)<=0.05*(1:m)/m) # k = k-hat
R<-(1:m)[p<=0.05*k/m]
```

There are many p -value-based methods to address this kind of multiple-testing problem. For instance, one may cite the modified Berk–Jones procedure.